

OSI Reference Model

- ➔ The OSI (Open System Interconnection) is conceptual model developed by the International Organizations of Standardization (ISO) in the year of 1984.
- ➔ This OSI Reference model provides some standard mechanism how data can be transmitted from one computer to another computer over the internet.
- ➔ The OSI model can divide the entire communication between one device to another device into **Seven** different layers, They are (Top to Bottom View)
 1. Application Layer
 2. Presentation Layer
 3. Session Layer
 4. Transport Layer
 5. Network Layer
 6. Data Link Layer
 7. Physical Layer

1. Applications Layer

- ➔ Application layer used by the network applications, Network applications means computer applications that use internet they are Google Chrome, Mozilla Firefox, Internet Explorer, Microsoft Edge, Outlook, Sky etc.
- ➔ All these applications are in the network using the application layer protocols to perform the functionalities
- ➔ All the network applications provide various services they are
 - File Transfer
 - Web Surfing/Browsing
 - Email
 - Virtual Terminal etc.
- ➔ In the Application layer file transfer is done with **FTP** protocol, Web browsing is done with the help of **HTTP/HTTPS**, for email using **SMTP** and for Virtual Terminal using **TELNET** protocols
- ➔ So, The Application layer provide different services for network applications with the help of protocols to perform the user activities.

2. Presentation Layer:

- ➔ Presentation Layer receives the data from application layer, the data in the form the **characters** and **numbers**.
- ➔ Presentation layer perform the following functions they are,
 - ❖ Data Translation.
 - ❖ Data Compression.

❖ Data Encryption and Decryption.

- Presentation layer translate the data (numbers/characters) into the machine understandable binary coded and this function of presentation layer is known as **Data Translation**

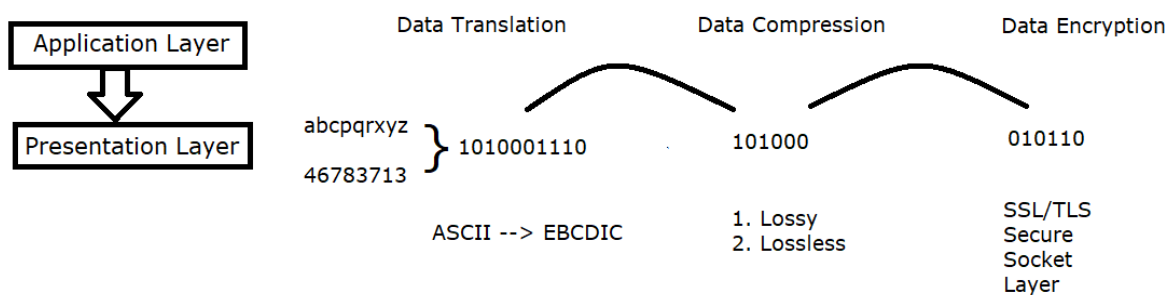
Ex: ASCII → EBCDIC

- Before data is transmitted, the presentation layer reduced the number of bits that are used to represent original data and this technique at presentation layer is called as **Data Compression Technique**

They are two types

1. Lossy
2. Lossless

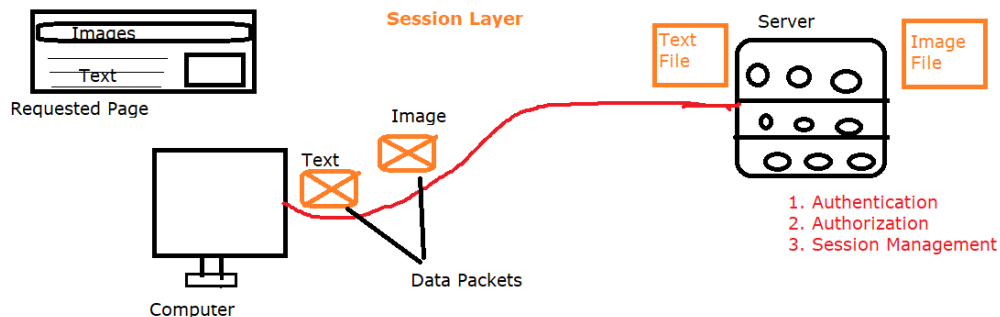
- Data compression technique is reduce the amount of space used to store the original file as the size of the file is reduced it can be received at destination in very less time. Also data compression technique most useful in real world for video and audio streaming.
- To maintain the integrity of the data, before transmitted the data is encrypted.
- We are using SSL/TLS protocol to perform the encryption and decryption to maintain the integrity of the data before transmitted.
- So, Presentation layer provide the following three functions on the data in the network
- ❖ Data Translation
 - ❖ Data Compression
 - ❖ Data Encryption



3.Session Layer:

- Session layer setting up many things connections for enabling the sending and receiving the data followed by connections and sessions terminations.
- Session layer has its own helpers known as API's and one of the API name is **NETBIOS** (Network Basic Input Output System) it allows application on different computers to communicate each other.

- ➔ Just before a session/connection is established with server, A server perform a function called as **Authentication**.
- ➔ **Authentication** is a process of verifying the person on the network who is legitimate user or not. For this purpose server use it user name and password once user entered username and password are match a session or connection is established between your computer and server.
- ➔ After authentication session layer perform authorization. **Authorization** is a function used by the server that determines whether you have a permission to access a file on the server. If not you will get a message like you are not authorized permission to access the file
- ➔ Session layer keeps a track of the files that are being downloaded. For example web page contains text, images etc, these text and images are stored in a separate files on the web server.
- ➔ When you request a website in a web browser your web browser opens a separate session to the web server to download each of these text files and images files separately. These files are received in the form the data packets. Session layer keeps a track of which data packet belongs to the which file is it text file or image file and tracks where the received data packet go in this case it goes to web browser that is session layer helps in **session management**.

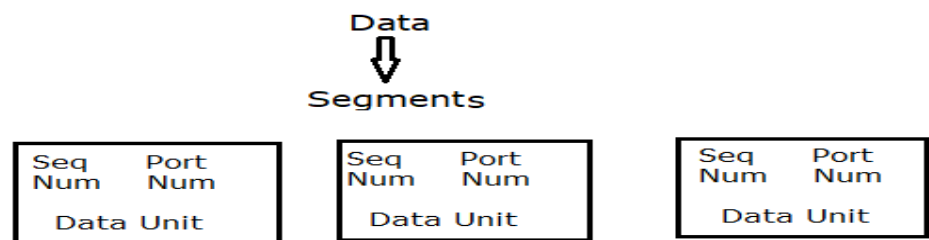


- ➔ Thus Session layer perform the following functions, They are
 1. Session Management
 2. Authentication
 3. Authorization

4. Transport Layer

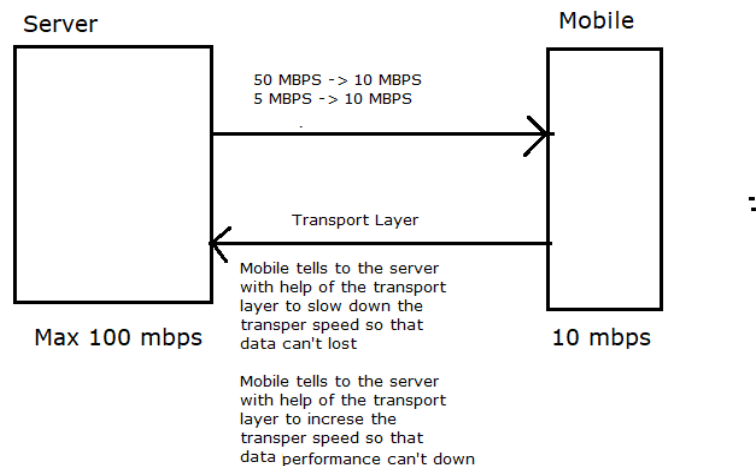
- The Transportation layer controls the reliability to segmentation, flow control and error control.
- In segmentation data received from session layer is divided into small data units are known as segment. Each segment contains a source and destination port number and a sequence number.
- Port number helps to direct each segment to the correct application and sequence number helps to reassemble the segment in correct order to forward the data in correct order to the receiver

Segmentation



Flow Control

- In flow control, Transport layer control the amount of data being transmitted.
Ex: --



Error Control:

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- ➔ In Error control, if some data does not arrive the destination, transport layer uses **Automatic Repeat Request (ARR)** to retransmit the lost or corrupted data. A group of bits called **Checksum** is added to each segment by the transport layer to find out receive corrupted segment

Protocols of Transport Layer are

- ❖ Transmission Control Protocol (**TCP**)
- ❖ User Datagram Protocol (**UDP**)

Transport layer perform other two types of services.

- ❖ Connection Oriented Transmission is done via **TCP**
- ❖ Connection less Oriented Transmission is done via **UDP**
- ➔ UDP is faster than TCP because it does not provide any feedback whether data was really delivered, where TCP provide a feedback therefore lost data can retransfer in TCP
- ➔ UDP is used only where does not a matter whether we have receive whole data
Ex: Online Streaming Movies, Songs, Games, Voice over IP etc.
- ➔ TCP is used only where whole data delivery is must
Ex: WWW, Email, FTP etc.
- ➔ So that transport layer involves in
 - ❖ Segmentation
 - ❖ Flow Controls
 - ❖ Error Controls
 - ❖ Connection Oriented Transmission
 - ❖ Connection less Transmission
- ➔ Transport layer using TCP and UDP Protocols to transform the data over internet

Differences Between TCP and UDP Protocols:

Key Differences	Transmission Control Protocol (TCP)	User Datagram Protocol (UDP)
Orientation	TCP is a reliable and Connection Oriented Protocol	UDP is a unreliable and Connection less Protocol
Faster	TCP slower than UDP to transmit the data	UDP is more faster than the TCP in transmit the data
Feedback	TCP provides feedback whether data is reach to the destination or not	UDP does not provides feedback whether data is reach to the destination or not
Usages	TCP used in www, Email and file transfer	UDP used in Online streaming movies, games, songs etc..

5. Network Layer

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- ➔ Transport layer passes segments to the network layer, where network layer works for the transmission of the received data segment from one computer to the another computer located in two different networks
- ➔ Data units in network layer are called as **Packets**
- ➔ The functions of the network layer are as follows
 - ❖ Logical Addressing
 - ❖ Routing
 - ❖ Path Determination

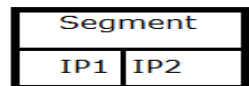
Logical Addressing:

- ➔ IP Addressing is done in network layer is called as logical addressing
- ➔ Every computer in a network has a unique IP address. Network layer assigns sender and receiver IP address to each segment to form a packet so that each data packet can reach correct destination.

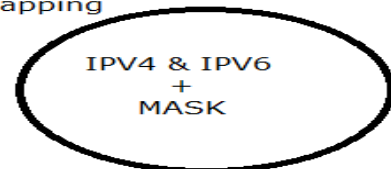
Routing:

- ➔ Routing is a method of moving data packet from source to destination and it is based on the logical addressing format (IPV4 & IPV6)

Packet Representaion in Network layer



Logical Address Mapping



Routing Process

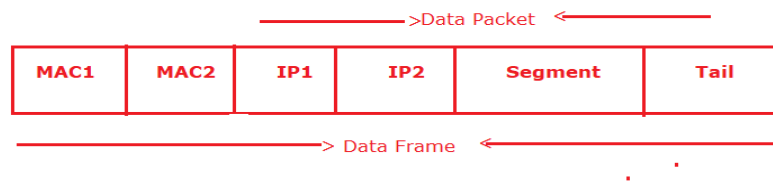
Path Determination:

- ➔ Generally a computer can be connected to the internet server or a computer in a number of ways choosing the best possible path for data delivery from source to destination is called path determination.
- ➔ Network layer using the following protocols to choose best path.
 - ❖ OSPF (Open shortest Path First)
 - ❖ BGP (Boarder Gateway Protocol)
 - ❖ IS-IS (Intermediate System to Intermediate System)

6. Data Link Layer:

- ➔ Data Link Layer receives the data packets from network layer.
- ➔ There are two types of addressing
 1. Logical Addressing
 2. Physical Addressing
- ➔ Each data segment contains sender IP address and receiver IP address to form a data packet is called logical addressing is done at network layer
- ➔ Data Link Layer use a functionality that is physical addressing, In physical addressing data link layer add MAC address of both sender and receiver to form a **Frame**
- ➔ **MAC** address is a 12 digit alphanumeric number which is printed on Network Interface Card (NIC)
- ➔ **Data Unit** in Data Link Layer is called Frame

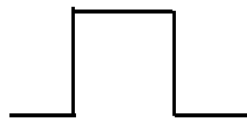
Frame Representation:



- ➔ Data link layer is embedded as software on the Network Interface Card (**NIC**) which transform the frame on the network from one computer to another computer over the **local media** such as Copper wire, Optical Fiber or Air
 - ➔ Data link layer perform two basic functions.
 - ❖ Framing (Access the media)
 - ❖ Media Access Control and Error Detection
- ➔ It controls how data is placed and received on media

7. Physical Layer:

- ➔ Physical Layer convert the sequence of bits into signal and transmit over the local media
- ➔ Signal can be a



Electrical Signal incase of Copper Cable or LAN Cable



Light Signal incase of Optical Fible



Radio signal incase of Air

